Skills Summary

Composite Area and Perimeter

Use this reference while completing your worksheet or when studying.

Rectangle Area

Rule: Area of a rectangle = length × width, in *square* units (cm^2, m^2, square units).

Example: A rectangle is 6 cm long and 5 cm wide. What is its area?
Step 1. Area is the space covered inside, so multiply the length by the width.
Step 2. $6 \times 5 = 30$.
Area: **30** $\mathbf{cm^2}$

Try it: A rectangle is 7 cm long and 3 cm wide. What is its area?

check: **21** cm^2

Rectangle Perimeter

Rule: Perimeter = add *all four* sides: $P = 2 \times (\text{length} + \text{width})$, in plain length units.

Example: A rectangle is 8 cm long and 2 cm wide. What is its perimeter?
Step 1. Perimeter is the distance around the edge, so add two lengths and two widths.
Step 2. $2 \times (8 + 2) = 2 \times 10 = 20$.
Perimeter: **20** cm

Try it: A rectangle is 6 m long and 5 m wide. What is its perimeter?

check: **22** m

Composite Area: Split, Then Add

Rule: Split an L-shape (or any rectilinear shape) into rectangles with a straight cut. Find each rectangle's area, then **add** the pieces.

Example: An L-shape splits into a bottom rectangle 6 units by 2 units and a top rectangle 3 units by 2 units. What is its area?
Step 1. The shape is not one rectangle, so cut it into two rectangles and add their areas.
Step 2. Bottom: $6 \times 2 = 12$. Top: $3 \times 2 = 6$. Add: $12 + 6 = 18$.
Area: **18** square units

Try it: An L-shape splits into a rectangle 5 units by 2 units and a rectangle 2 units by 3 units. What is its area?

check: **16** square units

Composite Perimeter: Walk Every Side

Rule: Walk all the way around and **add every side**. If a side label is missing, find it first: opposite directions must match (the small tops must add up to the whole bottom, the small rights to the whole left).

Example: An L-shape has sides 6, 2, 3, 2, 3, and 4 cm. What is its perimeter?
Step 1. Perimeter needs every side of the walk around the shape, so add all six labels — none may be skipped.
Step 2. $6 + 2 + 3 + 2 + 3 + 4 = 20$.
Perimeter: **20** cm

Try it: An L-shape has sides 8, 3, 4, 2, 4, and 5 cm. What is its perimeter?

check: **26** cm